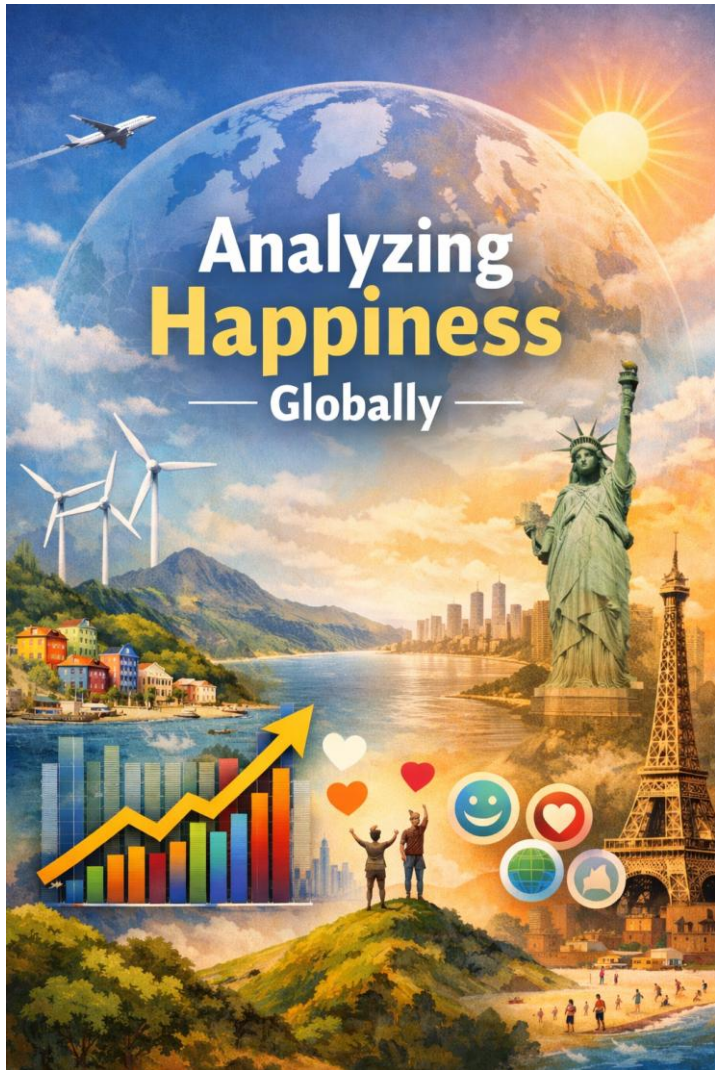




ABSTRACT

This study analyzes the 2019 World Happiness data using multivariate statistical techniques to uncover the key drivers of global well-being. After detecting and removing outliers, descriptive analysis and correlation assessment are conducted.

Factor analysis identifies two latent dimensions (Wealth & Life Quality and Freedom & Social Capital) which are then used in a regression model to explain happiness scores.

Finally, hierarchical and k-means clustering classify countries into well-being groups, highlighting clear global patterns in happiness levels.

AUTHORS

Nada Mohamed Abbas – 5220276
Shrouk Ekramy Anter – 5220126
Nada Hussain Abdullah – 5220853
Yomna Mukhtar Ali – 5220337
Nancy Yasser Mahmoud – 5220032

Course: Multivariate Data Analysis

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Professor: Marwah Sabry Siam

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Introduction

Happiness has become an important topic in social science research, economics, and public policy. For many years, countries focused mainly on economic indicators such as Gross Domestic Product (GDP) to measure development and progress. However, economic growth alone does not fully explain how satisfied people are with their lives. As a result, researchers and international organizations have started to study happiness as a broader measure of well-being that includes social, health, political, and psychological factors.

The World Happiness is one of the most well-known global efforts to measure and compare happiness across countries. It provides a happiness score for each country and examines the main factors that contribute to people's life satisfaction. This dataset allows researchers to understand why some countries are happier than others and which conditions help improve the quality of life for individuals.

This report uses the World Happiness Report 2019 dataset, which contains data for many countries around the world. The dataset is multivariate, meaning it includes several variables that jointly explain differences in happiness levels between countries. By using multivariate analysis, we can study how multiple factors interact and influence national happiness scores.

Background of the Material

The World Happiness dataset is based on survey data collected by the Gallup World Poll, combined with economic and health indicators from international organizations such as the World Bank and the World Health Organization (WHO). The main outcome of interest is the Happiness Score, which represents how people evaluate their overall life satisfaction.

Respondents are asked to imagine a ladder with steps numbered from 0 to 10, where 0 represents the worst possible life and 10 represents the best possible life. Each country's happiness score is calculated as the national average of these responses.

In addition to the happiness score, the dataset includes several explanatory variables that represent key dimensions of well-being, such as income, social support, health, freedom, generosity, and perceptions of corruption. These variables are widely supported in economic and psychological theories of happiness.

Importance of the Analysis

Studying the world happiness score is important for several reasons:

1. Beyond economic growth: Happiness provides a more complete picture of well-being than income alone. Countries with similar GDP levels can have very different happiness scores.

2. Policy implications: Understanding the factors that affect happiness helps governments design better social, economic, and health policies that improve people's quality of life.
3. Social and psychological well-being: Happiness is linked to better mental health, stronger social relationships, and higher productivity.
4. Global comparison: The dataset allows researchers to compare countries and regions, identify patterns, and learn from countries with high happiness levels.
5. Multidimensional analysis: Happiness is influenced by many interconnected factors, making it a suitable topic for multivariate statistical analysis.

Key Research Questions

The analysis seeks to address the following questions:

1. What are the underlying structural dimensions that explain national happiness?
2. How do these latent factors predict overall happiness scores?
3. How do countries naturally cluster based on their happiness profiles?

Methodology and analysis

Description of the data

Variable Name	Role	Type	Description
Happiness Score	Dependent (Y)	Quantitative (Continuous)	Represents the average life satisfaction of people in each country, measured on a scale from 0 to 10. It reflects how individuals evaluate their overall quality of life.
GDP per Capita (Log GDP per Capita)	Independent	Quantitative (Continuous)	Measures economic prosperity using the natural logarithm of GDP per capita adjusted for purchasing power parity (PPP), reflecting diminishing returns of income on happiness.
Social Support	Independent	Quantitative (Continuous, 0–1)	Indicates the proportion of people who have friends or relatives to rely on during difficult times; higher values imply stronger social networks.
Healthy Life Expectancy	Independent	Quantitative (Continuous)	Measures the expected number of years a person lives in good health, accounting for disability and illness.

Freedom to Make Life Choices	Independent	Quantitative (Continuous, 0–1)	Captures individuals' satisfaction with their freedom to make personal, economic, and social life choices.
Generosity	Independent	Quantitative (Continuous)	Reflects charitable behavior after adjusting for income; positive values indicate greater generosity than expected given income levels.
Perceptions of Corruption	Independent	Quantitative (Continuous, 0–1)	Measures the extent to which corruption is perceived in government and business; higher values indicate lower perceived corruption.

Note:

Overall rank: This variable is ordinal as it indicates the ranking position of each country in the happiness index based on its Happiness Score (with 1 being the highest score). Although it is related to the Happiness Score, it is a derived ordinal variable rather than an original explanatory measure — it simply orders countries by score and does not represent an independent factor influencing happiness itself, so it is excluded from the analysis of predictors.

Country or region: This is a categorical identifier variable listing the name of each country in the dataset. It does not measure a numerical attribute related to happiness determinants, and its purpose is solely to label observations (countries), so it is excluded from numerical analyses like correlation and regression.

Data cleaning

Data cleaning is an essential step in ensuring the quality and reliability of the analysis. It involves checking for missing values, inconsistencies, and outliers that may distort results. In this study, outlier detection methods were applied to identify extreme observations, and only a small number of potential outliers were found, indicating that the dataset is generally well-behaved. As a result, the cleaned data provides a more accurate and robust basis for analyzing the factors influencing world happiness scores.

The weighted BACON algorithm successfully identified the robust center and covariance structure of the data, converging after 6 iterations at a significance level of $\alpha = 0.05$. The analysis detected 2 potential outliers, representing 1.28% of the observations, indicating that the dataset is largely homogeneous with only a very small number of extreme cases. This suggests that the overall results are not strongly influenced by outliers, and the data structure can be considered stable for further statistical analysis.

Descriptive analysis:

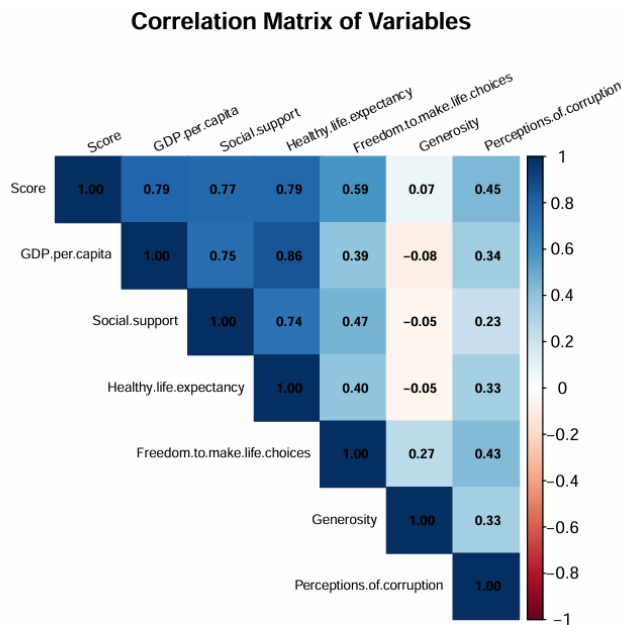
Before proceeding to advanced statistical analysis, an exploratory data analysis is conducted to understand the distribution of the variables. This step includes calculating summary statistics such as means and standard deviations, followed by the use of histograms to visually examine the shape, spread, and skewness of the data, as well as to identify potential outliers.

A. Summary Statistics

Variable	N	Mean	SD	Median	Min	Max	Range	Skewness	Kurtosis
Happiness Score	154	5.43	1.10	5.41	2.85	7.77	4.92	0.01	-0.63
GDP per Capita	154	0.91	0.40	0.97	0.00	1.68	1.68	-0.40	-0.79
Social Support	154	1.21	0.30	1.28	0.00	1.62	1.62	-1.14	1.20
Healthy Life Expectancy	154	0.73	0.24	0.79	0.10	1.14	1.04	-0.54	-0.59
Freedom to Make Life Choices	154	0.39	0.14	0.42	0.00	0.63	0.63	-0.68	-0.14
Generosity	154	0.19	0.10	0.18	0.00	0.57	0.57	0.73	1.04
Perceptions of Corruption	154	0.11	0.09	0.09	0.00	0.45	0.45	1.62	2.33

The descriptive statistics summarize data for 154 countries. The average happiness score is 5.43 with moderate variation ($SD = 1.10$) and an approximately normal distribution, making it suitable for further multivariate analysis. GDP per capita, social support, healthy life expectancy, and freedom show moderate dispersion and are mostly negatively skewed, indicating that many countries score relatively high on these dimensions. In contrast, generosity and perceptions of corruption are positively skewed, meaning most countries have low values with a few extreme cases. Overall, the variables show sufficient variability and acceptable distributional properties to analyze the determinants of national happiness.

B. Correlation matrix of Variables



The correlation matrix indicates that the life happiness score is strongly and positively associated with GDP per capita, healthy life expectancy, and social support ($r \approx 0.77\text{--}0.79$), suggesting that higher income, better health, and stronger social networks are key contributors to happiness. Freedom to make life choices shows a moderate positive correlation with happiness ($r \approx 0.59$), while perceptions of corruption are also moderately related ($r \approx 0.45$). In contrast, generosity has a very weak correlation with the happiness score ($r \approx 0.07$), indicating a limited direct relationship. Overall, the results emphasize the dominant role of economic, health, and social factors in explaining variations in life happiness across countries.

C. Ranking countries

Top 5 countries:

Country	Score	GDP per Capita	Social Support	Healthy Life Expectancy	Freedom to Make Life Choices	Generosity	Perceptions of Corruption
Finland	7.769	1.340	1.587	0.986	0.596	0.153	0.393
Denmark	7.600	1.383	1.573	0.996	0.592	0.252	0.410
Norway	7.554	1.488	1.582	1.028	0.603	0.271	0.341
Iceland	7.494	1.380	1.624	1.026	0.591	0.354	0.118
Netherlands	7.488	1.396	1.522	0.999	0.557	0.322	0.298

After cleaning the data, the results show that Finland ranks first with the highest happiness score (7.769), followed by Denmark (7.600) and Norway (7.554). Iceland (7.494) and the Netherlands (7.488) complete the top five, indicating that Nordic countries consistently dominate the highest happiness rankings, reflecting strong economic conditions, social support systems, and high quality of life.

Bottom 5 countries:

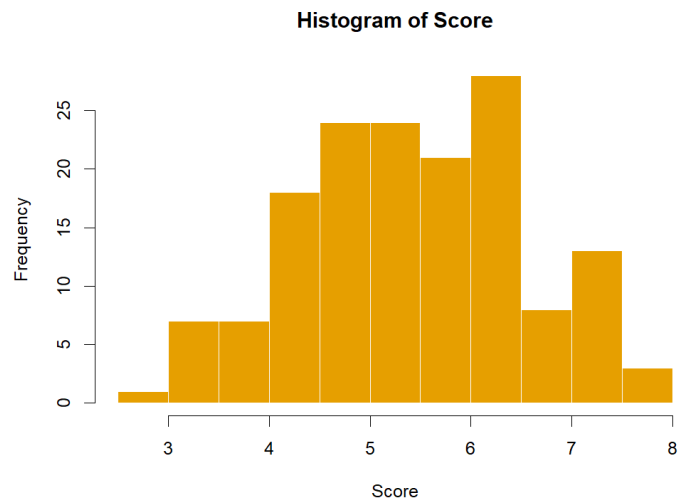
Country	Score	GDP per Capita	Social Support	Healthy Life Expectancy	Freedom to Make Life Choices	Generosity	Perceptions of Corruption
South Sudan	2.853	0.306	0.575	0.295	0.010	0.202	0.091
Central African Republic	3.083	0.026	0.000	0.105	0.225	0.235	0.035
Afghanistan	3.203	0.350	0.517	0.361	0.000	0.158	0.025
Tanzania	3.231	0.476	0.885	0.499	0.417	0.276	0.147
Yemen	3.380	0.287	1.163	0.463	0.143	0.108	0.077

The bottom five countries by happiness score are South Sudan, Central African Republic, Afghanistan, Tanzania, and Yemen, with scores ranging from 2.85 to 3.38, placing them at the lowest global ranks. This indicates very low reported life satisfaction, which is likely associated with factors such as economic hardship, political instability, conflict, weak social support, and limited access to healthcare and basic services.

D. Graphical Representation of Data

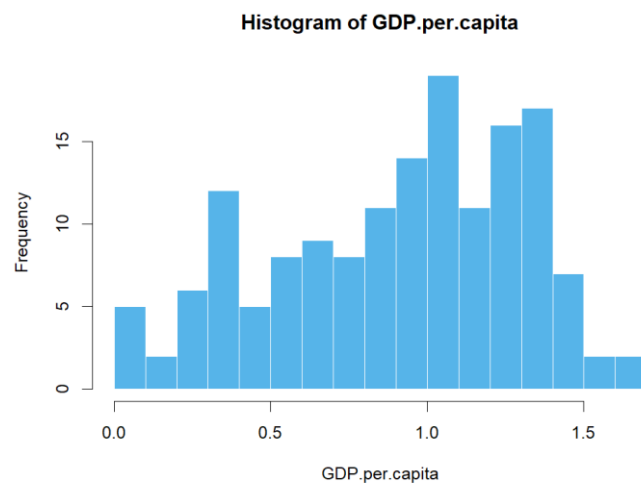
In parallel with statistical summaries, the data will be explored through a series of visualizations aimed at uncovering patterns and trends that may not be immediately evident through numerical metrics alone. This includes the use of histograms, density plots, boxplots, and bar charts to visualize the distribution of variables and identify potential outliers, skewness, and subgroup differences.

1) Histogram of happiness score



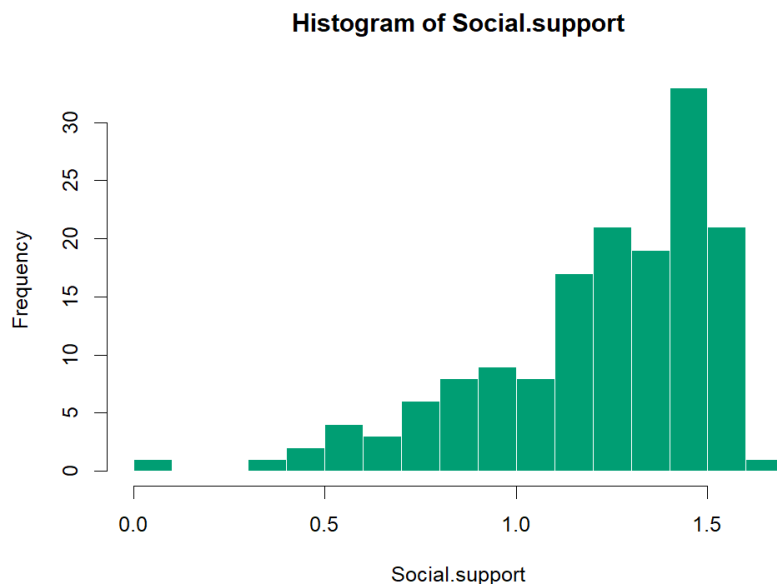
The histogram shows the distribution of the happiness score (dependent variable) across countries. Most countries have happiness scores between 4 and 6.5, indicating that the majority report moderate levels of life satisfaction. The distribution appears approximately symmetric and close to normal, with no extreme outliers. A small number of countries fall at the lower end (below 3.5) and the higher end (above 7), reflecting differences in national well-being. Overall, the shape of the histogram suggests that the happiness score is well distributed and suitable for further statistical and multivariate analysis.

2) Histogram of GDP per capita



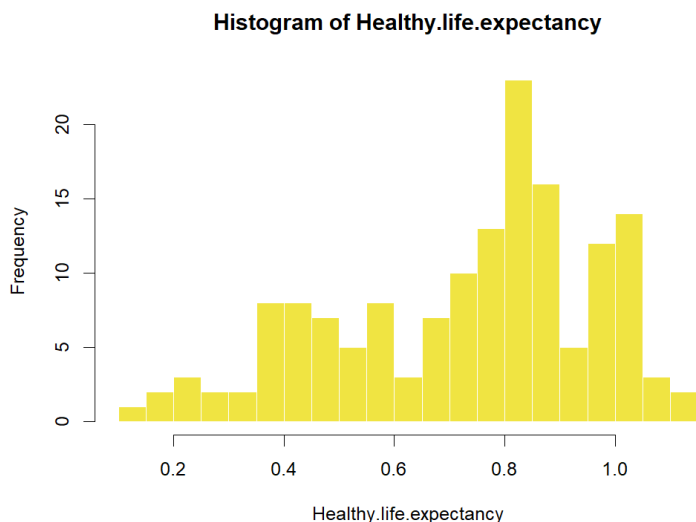
The histogram shows that GDP per capita is unevenly distributed across countries, with most observations concentrated in the mid-range and fewer countries at very low or very high levels. This variation suggests that GDP per capita may be an important factor influencing differences in life happiness scores, though it is unlikely to be the only determinant.

3) Histogram of social support



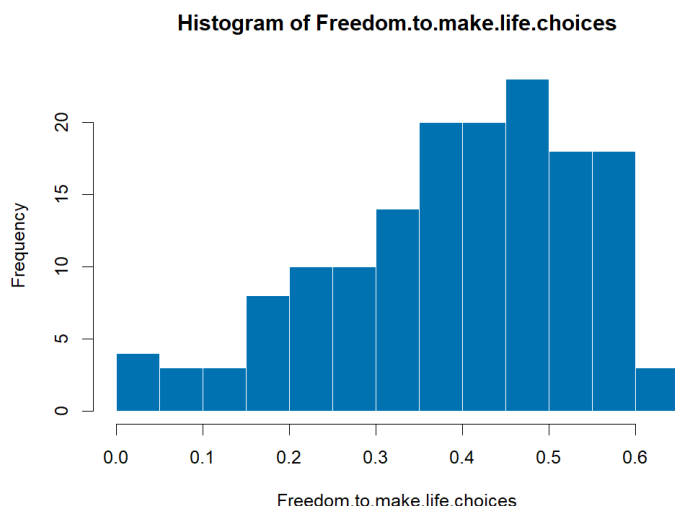
The histogram shows that social support is skewed toward higher values, with a large proportion of countries reporting strong levels of social support and only a small number with low support. This indicates that social relationships and community support are widely present across countries. Given the well-established link between social support and well-being, this distribution suggests that social support is likely a significant factor contributing to variations in life happiness scores, alongside economic and institutional factors.

4) Histogram of healthy life expectancy



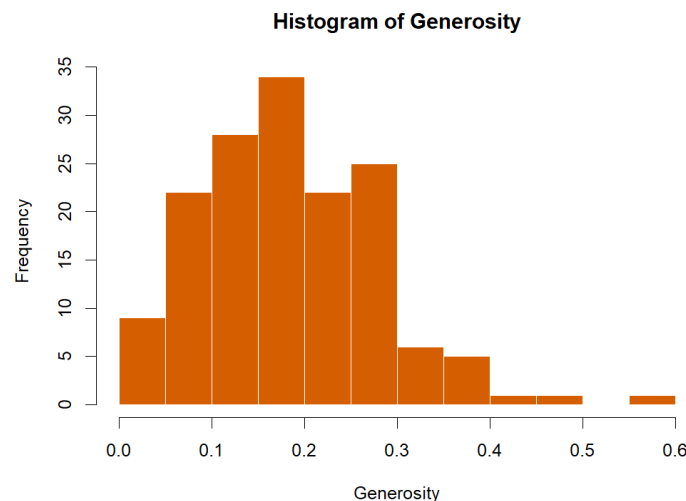
The histogram shows that healthy life expectancy is mostly concentrated at moderate to high values, with fewer countries experiencing very low healthy life expectancy. This indicates notable differences in health outcomes across countries, likely reflecting variations in healthcare quality, living conditions, and public health systems. Since good health is a key component of overall well-being, this distribution suggests that healthy life expectancy plays an important role in explaining differences in life happiness scores among countries.

5) Histogram of freedom to make life choices



The histogram shows that most countries are concentrated around moderate to relatively high levels of freedom to make life choices, with values mainly between 0.3 and 0.55. Very low values (below 0.1) are rare, indicating that only a small number of countries experience extremely limited personal freedom. The distribution is slightly left-skewed, suggesting that while high freedom is common, a minority of countries lag significantly behind. Overall, the plot indicates that freedom of choice is generally prevalent worldwide and likely plays an important role in explaining variations in the World Happiness

6) Histogram of generosity to make life choices



The histogram of Generosity shows a right-skewed distribution, indicating that most countries have low to moderate generosity scores, concentrated around 0.15–0.2. Scores range from 0 to 0.6, with only a few countries exhibiting high generosity above 0.4, which can be considered outliers. Overall, the majority of countries demonstrate modest levels of generosity, while very high generosity is relatively rare.

Factor Analysis

Before extracting the factors, the suitability of the data for factor analysis was assessed.

A. Check Suitability

1) Kaiser-Meyer-Olkin

The **Kaiser-Meyer-Olkin (KMO)** measure was **(0.75)**, which is above the recommended value of **0.60**, indicating that the sample size is **adequate** for factor analysis.

The Variables	Value of KMO	Interpretation
Social support	0.82	Social support (0.82) showed a strong level of adequacy , indicating it is highly suitable for the factor model.
Freedom to make life choices	0.78	Represents a Substantial level of adequacy.
Healthy life expectancy	0.76	Indicates a Reliable level of adequacy.
GDP per capita	0.73	Reflects a Good/Acceptable level of sampling adequacy.
Perceptions of corruption	0.71	Reaches an Adequate level of sampling adequacy. This confirms it is a viable component for the model despite its lower correlation compared to economic indicators .
Generosity	0.56	Generosity (0.56) had the lowest MSA value . While it still meets the minimum requirement (> 0.50) to be included in the analysis, it shows the weakest correlation with the other variables in the set .

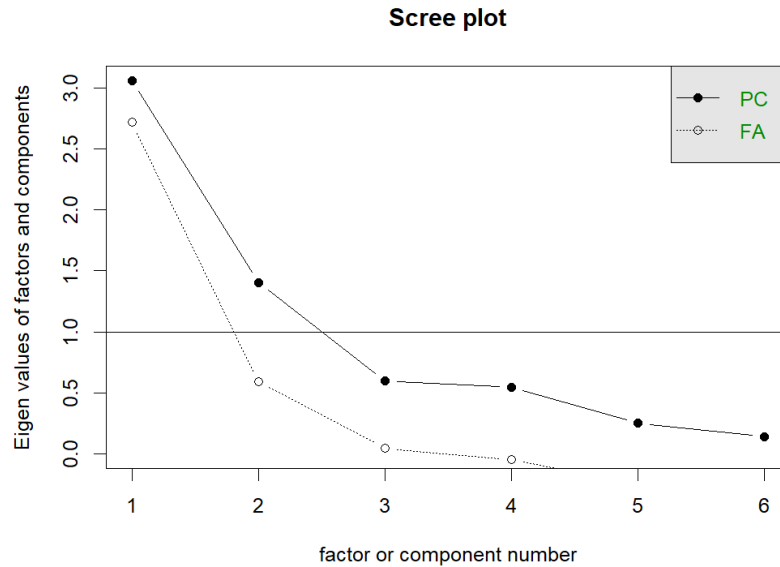
2) Bartlett's Test of Sphericity

H₀: The correlation matrix is an **identity matrix**

H₁: The correlation matrix is **not an identity matrix**

The results of **Bartlett's Test of Sphericity** yielded a p-value of **less than 0.05** (p-value = **1.714924e-86**). Therefore, we reject the null hypothesis(**H₀**), which states that the correlation matrix is an identity matrix. This rejection indicates that there are significant correlations between the variables in the dataset, confirming that the data is suitable and appropriate for **Factor Analysis**.

3) Scree Plot



To determine the optimal **number of factors**, the **Scree Plot** was examined, revealing an elbow after the initial factors where the curve levels off. Principal Component Analysis (**PC**) suggested retaining **two factors**, whereas Factor Analysis (**FA**) supported the retention of **one factor**.

4) Multivariate Normality Assessment

H0: The data follows a multivariate normal distribution.

H1: The data do not follow a multivariate normal distribution.

The Royston test was applied to assess multivariate normality for the dataset. The test statistic was 124.077 with a p-value < 0.001 , indicating that the null hypothesis of multivariate normality is rejected. This means the data do not follow a multivariate normal distribution, which suggests that methods requiring this assumption, such as Maximum Likelihood (ML), may not be appropriate. Instead, factor analysis methods that do not rely on multivariate normality, such as Principal Axis (PA) or Minimum Chi-square (MinChi), are more suitable for this dataset.

B. Factors without Rotation

1) 1 Factor with PA method

- Hypotheses Testing**

H₀: 1 Factor is sufficient

H₁: 1 Factor is insufficient

The **Chi-Square** test resulted in a value of **69.4** with a p-value (**p = 2e-11**). Since the p-value is significantly **less than 0.01**, we **reject** the null hypothesis, suggesting that a **one-factor** model **may not be perfectly sufficient** to explain the correlations among variables.

- **Model Reliability and Fit Indices**

Tests	Values of Tests	Interpretation
Tucker Lewis Index (TLI)	0.768	The value is 0.768 , which is below the ideal threshold of 0.90 , indicating a moderate fit.
RMSEA Index	0.209	The value is 0.209 , is higher than 0.10 and suggests a poor fit, reinforcing the possibility that more factors might be needed.
RMSA (Root Mean Square of Residuals)	0.13	The value is 0.13 , representing the average residual correlation.

- **Factor Extraction**

Based on Kaiser's Guttman (Eigenvalues > 1), the results show two eigenvalues greater than 1 (3.059 and 1.405). This suggests that extracting two factors would be more appropriate than one. However, in the current one-factor solution (PA1), the extracted factor explains 45.3% of the total variance (Proportion Var = 0.453).

- **Factor Loadings**

- The variables **GDP per capita** (0.904), **Healthy life expectancy** (0.895), and **social support** (0.814) show **very high** loadings on this factor.
- **Freedom to make life choices** (0.522) and **Perceptions of corruption** (0.403) show moderate loadings.
- **Generosity** has a negligible loading, indicating it does not contribute to this factor.

- **Communalities (h^2)**

The communalities represent the proportion of each variable's variance explained by the factor. GDP per capita (0.817) and Healthy life expectancy (0.800) are well-represented in the solution. Conversely, Generosity has a very low communality (0.001), suggesting that the current factor model fails to explain its variance.

2) 2 Factors with PA method

- **Hypotheses Testing**

H₀: 2 Factors are sufficient

H₁: 2 Factors are insufficient

The **Chi-Square** value is **13.86** with a **p-value of 0.0077**. Although the **p-value** is still **less than 0.01**, it is a **significant** improvement from the **one factor**.

- **Model Reliability and Fit Indices**

Tests	Value of Tests	Interpretation
Tucker Lewis Index (TLI)	0.914	The value improved to 0.914 , which exceeded the 0.90 threshold, indicating a good model fit .
RMSEA Index	0.126	The value is 0.126 . While still slightly high , the lower bound of the 90% confidence interval (0.058) suggests a much better fit than the previous model .
BIC	-6.28	Since the BIC is now negative , this provides evidence that the two-factor model is more parsimonious and better suited for the data.

- **Factor Extraction**

Consistent with Kaiser's Guttman, two factors with eigenvalues greater than 1 were retained (3.059 and 1.405). Together, these two factors explain 60.7% of the total variance (Cumulative Var = 0.607), which is a substantial increase from the 45.3% explained by the single factor model.

- **Factor Loadings**

- **Factor 1 (PA1):** Shows strong loadings for GDP per capita (0.907), Healthy life expectancy (0.891), and social support (0.805). This factor can be interpreted as "Socio-Economic Infrastructure".
- **Factor 2 (PA2):** Shows a high loading for Generosity (0.621) and a moderate loading for Perceptions of corruption (0.491). This factor reflects "Social Values and Governance".
- The variable "Perceptions of corruption" and "Freedom to make life choices" show cross-loadings or moderate contributions, indicating their complex nature.

- **Communalities (h^2)**

The communalities have significantly improved in this model. For instance, Generosity, which had a communality of 0.001 in the one-factor model, now has a communality of 0.388. The highest representation is for GDP per capita (0.864) and Healthy life expectancy (0.825), indicating that the two-factor solution captures the essence of these variables effectively.

3) 2 Factors with minchi method

- Hypotheses Testing**

H₀: 2 Factors are sufficient

H₁: 2 Factors are insufficient

Using a significance level of $\alpha = 0.01$, the chi-square test produced a p-value of 0.011, which is slightly greater than the chosen alpha level. Therefore, the null hypothesis is not rejected at the 1% significance level. This indicates that, although the model does not achieve perfect fit.

- Improved Fit Indices**

Tests	Values of Tests	Interpretation
TLI (Tucker Lewis Index)	0.921	It increased to 0.921 (from 0.768 in the one-factor model). Since it is now above 0.90 , we can conclude that the model has a Good Fit .
RMSEA	0.121	The value improved to 0.121 , and the lower bound of its confidence interval (0.053) suggests an acceptable fit .
BIC	-7.03	The BIC dropped to -7.03 . A lower (and specifically negative) BIC indicates a better and more parsimonious model .

- Factor Extraction**

Consistent with Kaiser's Guttman criterion, two factors with eigenvalues greater than 1 were retained (3.058 and 1.405), which are nearly identical to the PA solution (3.059 and 1.405). Together, these two factors explain **60.7% of the total variance** (Cumulative Var = 0.607), the same as in PA.

- Factor Loadings**

- **Factor 1 (MC1):** Shows strong loadings for **GDP per capita (0.933)**, **Healthy life expectancy (0.915)**, and **Social support (0.808)**. Compared with PA1 (GDP 0.907, Health 0.891, Social 0.805), minchi slightly increases the loadings, reinforcing the "**Socio-Economic Infrastructure**" factor.
- **Factor 2 (MC2):** Shows a high loading for **Generosity (0.629)** and **Perceptions of corruption (0.544)**. Compared with PA2 (Generosity 0.621, Corruption 0.491), minchi slightly strengthens both loadings, maintaining the "**Social Values and Governance**" interpretation.
- The variable **Freedom to make life choices** shows more balanced contributions across factors in minchi (0.471 on MC1 and 0.466 on MC2), whereas in PA it loaded mostly on PA1 (0.557). This highlights minchi's ability to capture its multidimensional nature more clearly.

- **Communalities (h^2)**

The communalities in minchi are very similar to PA:

- **GDP per capita:** 0.875 (PA: 0.864)
- **Healthy life expectancy:** 0.839 (PA: 0.826)
- **Social support:** 0.652 (PA: 0.671)
- **Freedom to make life choices:** 0.438 (PA: 0.452)
- **Generosity:** 0.397 (PA: 0.388)
- **Perceptions of corruption:** 0.441 (PA: 0.442)

Among the factor analysis solutions, the 2-factor model using the Minchi method was selected as the most suitable. This choice is justified because it produced a slightly better chi-square p-value (0.011), indicating a closer fit to the data, and slightly improved fit indices (TLI = 0.921, RMSEA = 0.121, BIC = -7.03) compared with the PA solution. Therefore, the subsequent detailed interpretation and discussion will focus on this Minchi 2-factor solution, as it provides the most reliable and interpretable representation of the underlying constructs. Factors with rotation

1) Varimax Rotation

The rotated factor solution using Minchi (Minimum Chi-square, MinChi) with Varimax rotation reveals a clear and interpretable two-factor structure.

Factor 1 (MC1) shows very strong loadings for GDP per capita (0.93), Healthy life expectancy (0.91), and Social support (0.80). These high loadings indicate that this factor primarily captures the material and structural conditions of life, reflecting economic prosperity, population health, and social stability. The magnitude of these loadings suggests that these variables contribute substantially to the underlying latent construct, justifying the interpretation of this factor as Wealth and Life Quality.

Factor 2 (MC2) is characterized by moderate to strong loadings on Generosity (0.61), Perceptions of corruption (0.60), and Freedom to make life choices (0.54). This pattern indicates that the second factor reflects institutional and social freedoms, ethical governance, and prosocial behavior. Accordingly, this factor can be interpreted as Freedom and Social Capital, capturing aspects of trust, autonomy, and civic values that influence subjective well-being beyond material conditions.

The Varimax rotation results in a simple structure, with variables loading strongly on one factor and showing minimal cross-loadings on the other, supporting good factor separation and interpretability. Together, the two factors explain approximately 60.7% of the total variance, with Factor 1 accounting for the larger share (42.9%) and Factor 2 contributing an additional 17.8%. The communalities further indicate that most variables are well represented by the two-factor solution, particularly GDP per capita and Healthy life expectancy, which exhibit very high shared variance with the retained factors.

2) *Oblimin Rotation*

The factor analysis with Oblimin rotation produced a clear two-factor solution while allowing the factors to correlate.

Factor 1 (MC1) has very high loadings on GDP per capita (0.94), Healthy life expectancy (0.92), and Social support (0.80). This shows that Factor 1 mainly represents economic conditions, health, and basic life quality. The strong loadings indicate that these variables are closely related and together form an important dimension of life well-being. Therefore, this factor can be labeled Wealth and Life Quality.

Factor 2 (MC2) shows moderate loadings on Generosity (0.65), Perceptions of corruption (0.58), and Freedom to make life choices (0.51). This suggests that Factor 2 reflects social and institutional aspects of well-being, such as trust, freedom, and prosocial behavior. This factor can be labeled Freedom and Social Capital.

Because Oblimin rotation allows factors to correlate, the results show a moderate positive correlation between the two factors ($r = 0.26$). This indicates that countries with better economic and life conditions also tend to have higher levels of social freedom and trust, although the two factors remain distinct. Together, the two factors explain about 59.5% of the total variance, with Factor 1 explaining the larger share. The communalities indicate that most variables are well explained by the two-factor model, especially GDP per capita and Healthy life expectancy.

Although both Varimax (orthogonal) and Oblimin (oblique) rotations produced very similar factor structures with comparable loadings and communalities, the Varimax rotation was chosen for further interpretation. This decision is based on its simplicity, as it produces uncorrelated factors and is easier to compute and interpret. Additionally, in the factor loadings matrix, a cutoff point of 0.4 was applied to highlight meaningful loadings, ensuring clarity in identifying the main contributors to each factor.

Regression analysis

After reducing the original set of life-condition indicators using factor analysis with Varimax rotation, two orthogonal factors were retained, representing **Wealth & Life Quality** and **Freedom & Social Capital**. These factor scores were subsequently used as independent variables in a multiple linear regression model to examine their influence on life happiness scores. Regression analysis was appropriate at this stage as it allows for the quantification and statistical testing of the relationship between the extracted latent dimensions and happiness, while minimizing multicollinearity due to the orthogonality of the rotated factors.

The estimated regression model is:

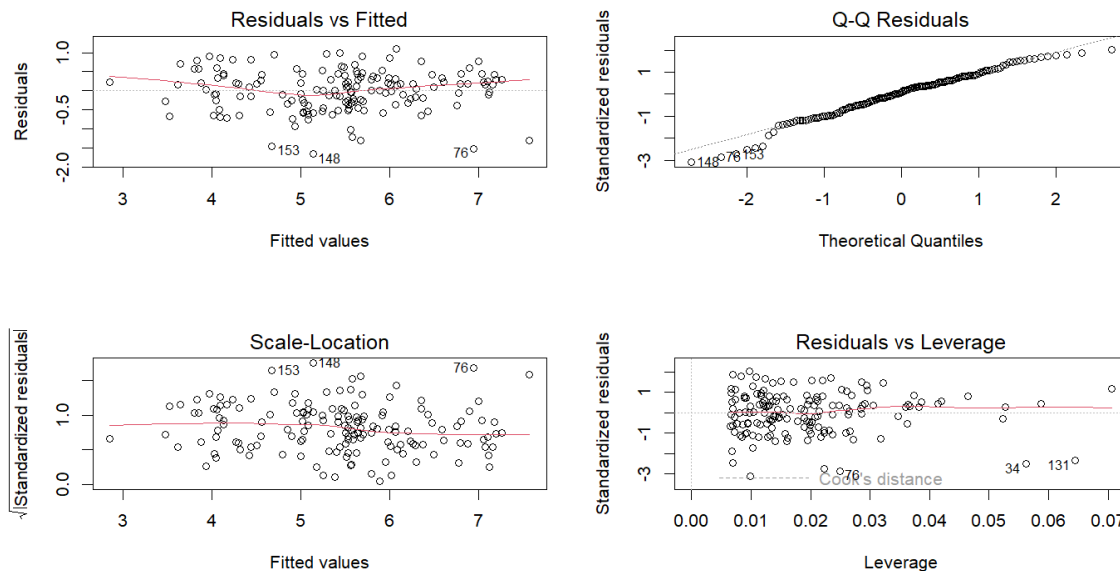
$$\hat{Y} = 5.428 + 0.915 \cdot \text{Wealth_Life_Quality} + 0.426 \cdot \text{Freedom_Social_Capital}$$

The results indicate that both factors have statistically significant and positive effects on happiness, with Wealth & Life Quality exerting the stronger influence. The model exhibits strong explanatory power, **explaining approximately 76.6%** of the variance in happiness scores, and the overall regression model is highly statistically significant.

A. Estimated Regression Model Interpretation

- **Intercept = 5.43:** Represents the expected life happiness score when both factor scores (**Wealth & Life Quality** and **Freedom & Social Capital**) are zero, i.e., when a country is at the average level of both factors.
- **Beta for Wealth & Life Quality = 0.92:** A one-unit increase in this factor is associated with an increase of approximately 0.92 points in the happiness score, holding the other factor constant.
- **Beta for Freedom & Social Capital = 0.43:** A one-unit increase in this factor leads to an increase of about 0.43 points in happiness, after controlling for Wealth & Life Quality.
- **Interpretation:** Improvements in both economic and social conditions raise happiness, with **Wealth & Life Quality** having the larger impact.

B. Testing Model Assumptions



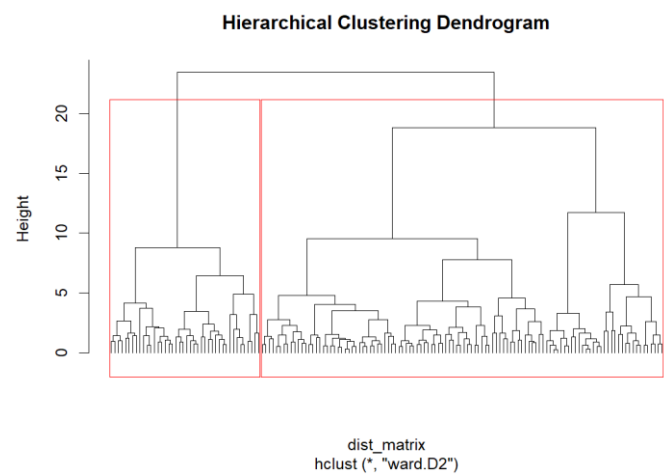
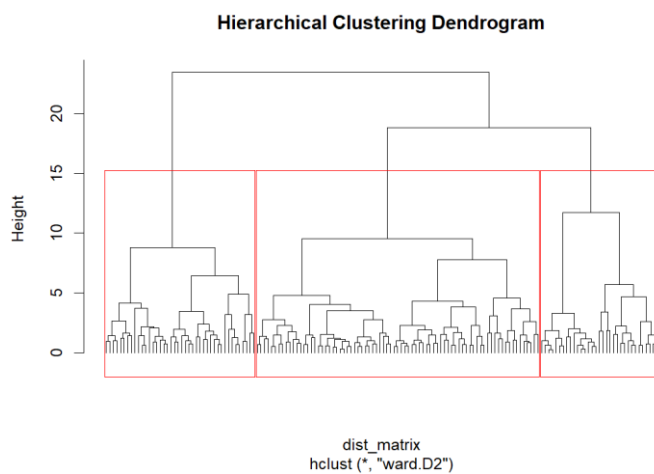
The assumptions underlying the multiple linear regression model were assessed using standard diagnostic procedures. Visual inspection of the residuals versus fitted values plot indicated an approximately random scatter around zero, supporting the assumption of linearity. The scale–location plot showed a relatively constant spread of standardized residuals across fitted values, suggesting that homoscedasticity was reasonably satisfied. Normality of residuals was evaluated using a normal Q–Q plot, which demonstrated that residuals largely followed the theoretical normal distribution, with only minor deviations in the tails. Independence of errors was considered acceptable given the cross-sectional nature of the data, and no influential observations were detected based on leverage and Cook’s distance diagnostics. Overall, no serious violations of regression assumptions were identified, indicating that the model is statistically reliable.

Cluster Analysis

A. Detection

1) Dendrogram

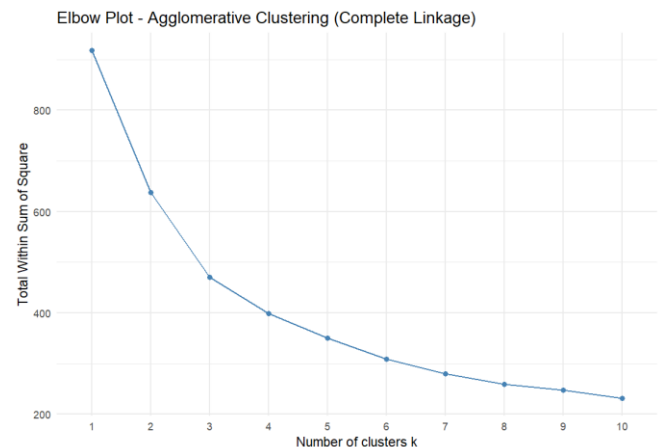
To apply Clustering Methods, we used Agglomerative Hierarchical Clustering Methods. Firstly, we scaled the variables because clustering methods are based on distance measures, which are heavily affected by the measurement unit. Euclidean distance was employed to compute the distance matrix between observations, and Ward's linkage method was used to merge clusters by minimizing the increase in ESS. This hierarchical approach was primarily performed to draw the dendrogram and determine the optimal number of clusters for further analysis.



The dendrogram suggested different possible numbers of clusters, and after examining the structure, we found that a three-cluster solution provided a more balanced distribution of observations across clusters. This approach ensures that each cluster contains a meaningful number of countries, improving interpretability and supporting further analysis with K-means clustering.

2) Elbow

The "elbow" appears around 2–3 clusters, indicating that 2 or 3 clusters is optimal. The plot shows a clear leveling off after 3 clusters, suggesting that adding more clusters beyond this point yields diminishing returns in reducing the within-cluster variance. Choosing 3 clusters might be slightly better for balancing cluster sizes, while 2 clusters captures the largest variance reduction but could merge moderately different groups.



B. K-means Clustering

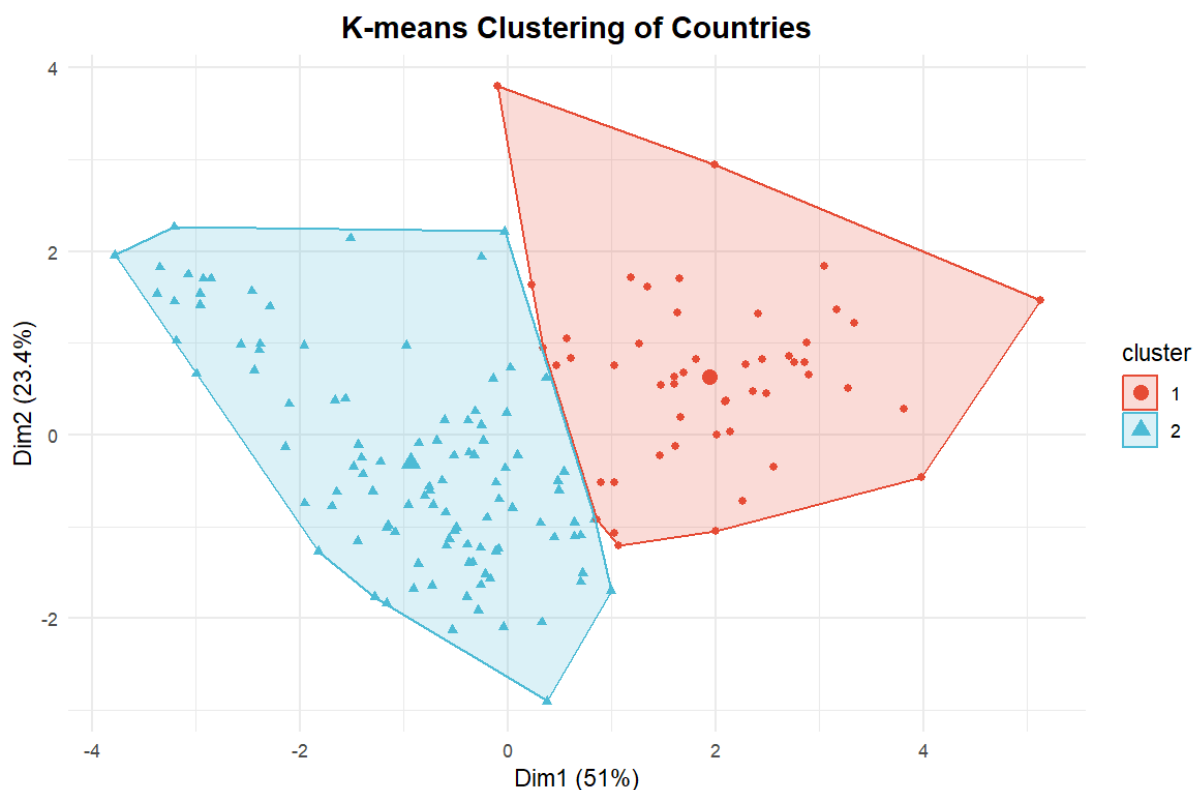
1) $K=2$

Based on the number of clusters determined from the hierarchical method using the dendrogram and Elbow plot, we started with **two** clusters in the non-hierarchical clustering to perform the K-means clustering.

Cluster	size	GDP per capita	Social support	Healthy life expectancy	Freedom to make life choices	Generosity	Perceptions of corruption
Higher Well-being	50	-1.1585967	-1.1068352	-1.1491528	-0.4897762	0.2209515	-0.16196184
Lower Well-being	104	0.5570177	0.5321323	0.5524773	0.2354693	-0.1062267	0.07786627

The results were: the first cluster contains 50 observations, has a within-cluster sum of squares of 183.4058, and is labeled Lower Well-being because mean values for GDP per capita (-1.15), Social support (-1.10), and Healthy life expectancy (-1.14) indicate that the 50 countries in this group face economic and health challenges. The second cluster contains 104 observations, has a within-cluster sum of squares of 423.4169, and is labeled High Well-being because mean values for GDP per capita (0.55), Social support (0.53), and Healthy life expectancy (0.55) reflect better living standards and stronger social systems for the 104 countries in this group.

Where within sum squares is 183.4058 in first cluster, 423.4169 in second cluster, these values are high and between sum squares / total sum squares is 33.9 % is low percentage.



The Factor Analysis reduced the data to two dimensions, capturing most of the variation between countries: Dim1 (51%) reflects differences in Wealth & Life Quality, while Dim2 (23.4%) represents Freedom & Social Capital.

The plot reveals two distinct clusters of countries:

- **Cluster 1 (red circles)** is positioned on the right side, with higher Dim1 values, indicating countries with greater wealth and higher life quality.
- **Cluster 2 (blue triangles)** appears on the left side, with lower Dim1 values, representing countries with lower wealth and life quality.

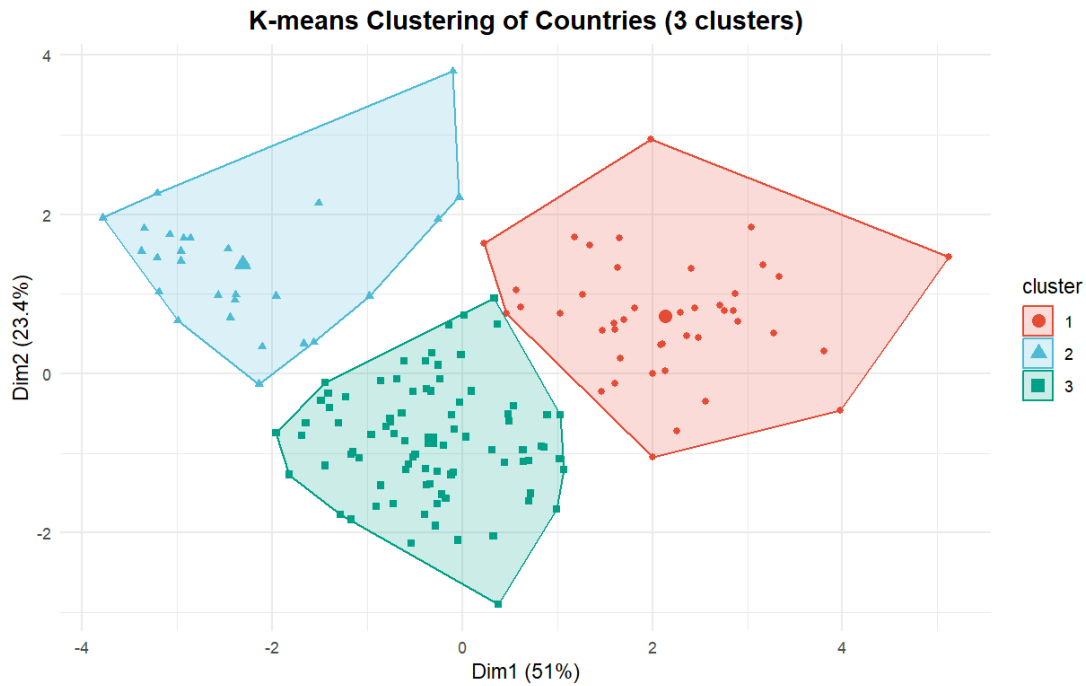
The separation between the clusters suggests a meaningful classification, showing that countries differ primarily in terms of wealth and life quality, with secondary differences in freedom and social capital.

2) $K=3$

K-means clustering was applied with k is 3, as indicated by the Elbow Method and Dendrogram.

Cluster	Size	GDP per capita	Social support	Healthy life expectancy	Freedom to make life choices	Generosity	Perceptions of corruption
Lower Well-being	43	-1.3016142	-1.1724411	-1.2871432	-0.5469405	0.2797086	-0.1722801
High Well-being	28	1.0300985	0.8279075	0.8826348	1.0429845	1.1902455	1.5279606
Moderate Well-being	83	0.3268271	0.3281152	0.3690769	-0.0684955	-0.5464378	-0.4262030

The results were: the first cluster contains 43 observations, has a within-cluster sum of squares of 131.82600, and is labeled Lower Well-being because it has the lowest mean values overall, with GDP per capita (-1.30) and Healthy life expectancy (-1.28), indicating major economic and health challenges. The second cluster contains 28 observations, has a within-cluster sum of squares of 91.64814, and is labeled High Well-being because it has the highest positive mean values, with Freedom to make life choices (1.04) and GDP per capita (1.03), reflecting the best living standards. The third cluster contains 83 observations, has a within-cluster sum of squares of 198.25344, and is labeled Medium Well-being because it has moderate values near the dataset average, e.g., GDP per capita (0.32) and Social support (0.32), providing a balanced classification for most countries.



The plot reveals **three distinct clusters** of countries:

- **Cluster 1 (red circles):** Positioned on the **right side** with higher Dim1 values, indicating countries with **greater wealth and higher life quality**. These countries are also relatively neutral on the Dim2 axis.
- **Cluster 2 (blue triangles):** Located on the **top-left**, with low Dim1 values but high Dim2 values. This represents countries with **lower wealth/life quality** but potentially higher levels of **freedom and social capital** relative to their economic status.
- **Cluster 3 (green squares):** Situated at the **bottom-center**, characterized by moderate to low Dim1 values and the **lowest Dim2 values**. These represent countries with lower levels of **freedom and social capital**, regardless of their wealth.

The separation between the clusters suggests a meaningful classification, showing that while **wealth and life quality** (Dim1) is the primary divider, there is a secondary, critical distinction in **freedom and social capital** (Dim2) that further differentiates countries into three groups rather than just two.

C. Optimal Cluster Selection

Based on the cluster detection results, a **three-cluster solution (K=3)** was selected as the optimal model for this analysis. The decision is justified by the following points:

- **Improved Balance:** Unlike the two-cluster model, which merged moderately different groups, the three-cluster solution provides a more balanced distribution of countries. It successfully isolates the "Moderate Well-being" group (83 countries), which represents the global average, leading to a more realistic classification.
- **Dimensional Clarity:** While Dimension 1 (Wealth & Life Quality) is the primary separator, the three-cluster approach captures the critical details of Dimension 2 (Freedom & Social Capital). This allows us to distinguish between countries that may have similar economic levels but differ significantly in social structures.
- **Methodological:** The choice of aligns with both the Dendrogram (showing clear sub-groupings) and the Elbow Method, where the within-cluster variance stabilizes. Beyond three clusters, the gains in statistical accuracy are minimal compared to the loss of simplicity.

Conclusion:

The analysis of the World Happiness Report 2019 dataset demonstrates that national happiness is influenced by multiple interconnected factors, including economic conditions, health, social support, freedom, and governance. Factor Analysis identified two key dimensions (Wealth & Life Quality and Freedom & Social Capital) which significantly explain variations in happiness, with regression results showing both positively impact life satisfaction, though economic and health factors have a stronger effect. Cluster analysis revealed three meaningful groups of countries (Lower Well-being, Medium Well-being, and High Well-being) highlighting differences not only in wealth and life quality but also in social freedom and capital. Overall, the findings underscore that happiness cannot be measured by GDP alone and that policies targeting both material conditions and social-institutional factors are essential to improving citizens' well-being across nations.

References

- Sustainable Development Solutions Network (SDSN). *World Happiness Report*. Kaggle **World Happiness** dataset, 2015–2019. Available at: <https://www.kaggle.com/datasets/unsdsn/world-happiness/data>

Appendix

```
# =====
# Load required libraries
# =====
library(tidyverse)
library(psych)      # Factor Analysis
library(factoextra) # Clustering visualization
library(VIM)        # kNN imputation if needed
library(wbacn)       # Outlier detection (BACON)
library(corrplot)
library(MVN)
# =====
# Load the dataset
# =====
df <- read.csv("2019.csv", stringsAsFactors = FALSE)

# =====
# Outlier detection (BACON)
# =====
w <- wBACON(df %>%
            select(-Overall.rank, -Country.or.region))
w
outlier_indices <- which(is_outlier(w))
# Remove outliers
df_clean <- df[-outlier_indices, ]

# =====
# DESCRIPTIVE ANALYSIS AFTER REMOVING OUTLIERS
# =====

cat("\n===== DESCRIPTIVE ANALYSIS (AFTER OUTLIER REMOVAL) =====\n")

# 1. Detailed descriptive statistics
cat("\n--- DETAILED DESCRIPTIVE STATISTICS ---\n")
desc_stats <- describe(df_clean %>% select(-Country.or.region, -Overall.rank))
print(desc_stats)

# 2. CORRELATION MATRIX
cat("\n--- CORRELATION MATRIX ---\n")
cor_matrix <- cor(df_clean %>% select(-Overall.rank, -Country.or.region))
print(round(cor_matrix, 3))
```

```

# Reduce top margin
corrplot(corr_matrix,
         method = "color",
         type = "upper",
         tl.col = "black",
         tl.srt = 25,
         tl.cex = 0.7,
         addCoef.col = "black",
         number.cex = 0.7,
         main = "Correlation Matrix of Variables",
         mar = c(0, 0, 1, 0))

# 3. Top and bottom 5 countries by Score
cat("\n--- TOP 5 COUNTRIES BY HAPPINESS SCORE ---\n")
top_5 <- df_clean %>%
  arrange(desc(Score)) %>%
  head(5)
print(top_5)

cat("\n--- BOTTOM 5 COUNTRIES BY HAPPINESS SCORE ---\n")
bottom_5 <- df_clean %>%
  arrange(Score) %>%
  head(5)
print(bottom_5)

#4. Histograms for all Variables

# Get numeric variables names only
numeric_vars <- df_clean %>%
  select(-Country, -region, -Overall.rank) %>%
  names()

# Histogram 1
hist(df_clean[[numeric_vars[1]]],
     main = paste("Histogram of", numeric_vars[1]),
     xlab = numeric_vars[1],
     col = "#E69F00",
     border = "white",
     breaks = 15)

# Histogram 2
hist(df_clean[[numeric_vars[2]]],
     main = paste("Histogram of", numeric_vars[2]),
     xlab = numeric_vars[2],
     col = "#56B4E9",
     border = "white",
     breaks = 15)

# Histogram 3
hist(df_clean[[numeric_vars[3]]],

```

```

    main = paste("Histogram of", numeric_vars[3]),
    xlab = numeric_vars[3],
    col = "#009E73",
    border = "white",
    breaks = 15)

# Histogram 4
hist(df_clean[[numeric_vars[4]]],
     main = paste("Histogram of", numeric_vars[4]),
     xlab = numeric_vars[4],
     col = "#F0E442",
     border = "white",
     breaks = 15)

# Histogram 5
hist(df_clean[[numeric_vars[5]]],
     main = paste("Histogram of", numeric_vars[5]),
     xlab = numeric_vars[5],
     col = "#0072B2",
     border = "white",
     breaks = 15)

# Histogram 6
hist(df_clean[[numeric_vars[6]]],
     main = paste("Histogram of", numeric_vars[6]),
     xlab = numeric_vars[6],
     col = "#D55E00",
     border = "white",
     breaks = 15)

# =====
# Keep numeric columns only and Exclude Y which is Score
# =====
df_numeric <- df_clean %>%
  select(-Overall.rank, -Country.or.region, -Score)

# =====
# Scale the data
# =====
df_scaled <- as.data.frame(scale(df_numeric))

# =====
# Factor Analysis - check suitability
# =====
KMO_result <- KMO(df_scaled)
print(KMO_result)

bartlett_result <- cortest.bartlett(cor(df_scaled), n = nrow(df_scaled))
print(bartlett_result)

```

```

# =====
# Determine number of factors
# =====

scree(df_scaled)                                # Scree plot

# =====
# Factor Analysis
# =====
mvn_result <- mvn(data = df_scaled, mvn_test = "royston")
mvn_result$multivariate_normality
mvn_result$univariate_normality
library(psych)

run_fa <- function(data, nfactors, rotate, fm, subtitle) {
  cat("\n===== ", subtitle, "=====\n")
  fa_res <- fa(data, nfactors = nfactors, rotate = rotate, fm = fm)

  # Print summary
  summary(fa_res)

  # Print loadings >= 0.4
  print(fa_res$loadings, cutoff = 0.4)

  # Print communalities
  cat("\nCommunalities:\n")
  print(fa_res$communality)

  # Print eigenvalues
  cat("\nEigenvalues:\n")
  print(fa_res$e.values)

  return(fa_res)
}

# =====
# Run Factor Analyses
# =====
fa_1 <- run_fa(df_scaled, nfactors = 1, rotate = "none", fm = "pa", "1 Factor, No Rotation")
fa_2 <- run_fa(df_scaled, nfactors = 2, rotate = "none", fm = "pa", "2 Factors, No Rotation")
fa_2_ml <- run_fa(df_scaled, nfactors = 2, rotate = "none", fm = "minchi", "2 Factors, minchi
Method, No Rotation")
fa_2_var <- run_fa(df_scaled, nfactors = 2, rotate = "varimax", fm = "minchi", "2 Factors,
minchi Method, Varimax Rotation")
fa_2_obl <- run_fa(df_scaled, nfactors = 2, rotate = "oblimin", fm = "minchi", "2 Factors,
minchi Method, Oblimin Rotation")

# =====
# Extract factor scores for regression
# =====

```

```

factor_scores <- as.data.frame(fa_2_var$scores)
# Rename the columns
colnames(factor_scores) <- c("Wealth_Life_Quality", "Freedom_Social_Capital")
# Append the Score column
factor_scores$Score <- df_clean$Score

# =====
# Multiple Linear Regression using factor scores
# =====
lm_model <- lm(Score ~ ., data = factor_scores)
summary(lm_model)

# =====
# Check diagnostics
# =====
par(mfrow=c(2,2))
plot(lm_model)
par(mfrow=c(1,1))

# =====
# Hierarchical clustering (Ward method)
# =====
dist_matrix <- dist(df_scaled, method = "euclidean")
hc <- hclust(dist_matrix, method = "ward.D2")

# Plot dendrogram
plot(hc, labels = FALSE, hang = -1, main = "Hierarchical Clustering Dendrogram")
rect.hclust(hc, k = 2, border = "red") # Cut dendrogram into 2 clusters

# =====
# Elbow method to determine optimal number of clusters
# =====

fviz_nbclust(df_scaled, FUN = hcut, method = "wss",
             hc_method = "ward.D",
             k.max = 10) +
  labs(title = "Elbow Plot - Agglomerative Clustering (Complete Linkage)") +
  theme_minimal()

# =====
# K-means clustering
# =====
set.seed(123)
k <- 2
kmeans_result <- kmeans(df_scaled, centers = k, nstart = 25)
kmeans_result
# Add cluster assignment to original data
df_clean$Cluster_1 <- kmeans_result$cluster
df_clean$Cluster_1 <- factor(df_clean$Cluster_1,

```

```

        levels = c(1, 2),
        labels = c("Lower Well-being",
                    "Higher Well-being"))

# =====
# Visualize K-means clusters
# =====
fviz_cluster(kmeans_result, data = df_scaled, geom = "point",
              ellipse.type = "convex", repel = TRUE,
              palette = c("#E64B35", "#4DBBD5")) +
  ggtitle("K-means Clustering of Countries") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5, size = 14, face = "bold"))

# =====
# Hierarchical clustering (Ward method) for 3 clusters
# =====
dist_matrix <- dist(df_scaled, method = "euclidean")
hc <- hclust(dist_matrix, method = "ward.D2")

# Plot dendrogram
plot(hc, labels = FALSE, hang = -1, main = "Hierarchical Clustering Dendrogram")
rect.hclust(hc, k = 3, border = "red") # Cut dendrogram into 3 clusters

# =====
# K-means clustering for k = 3
# =====
set.seed(123)
k <- 3
kmeans_result <- kmeans(df_scaled, centers = k, nstart = 25)
kmeans_result

# Add cluster assignment to original data
df_clean$Cluster_2 <- kmeans_result$cluster
df_clean$Cluster_2 <- factor(df_clean$Cluster_2,
                             levels = c(1, 2, 3),
                             labels = c("Lower Well-being",
                                          "Higher Well-being",
                                          "Medium Well-being"))

# =====
# Visualize K-means clusters (3 clusters)
# =====
fviz_cluster(kmeans_result, data = df_scaled, geom = "point",
              ellipse.type = "convex", repel = TRUE,
              palette = c("#E64B35", "#4DBBD5", "#00A087")) +
  ggtitle("K-means Clustering of Countries (3 clusters)") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5, size = 14, face = "bold"))

```